

The coherence indistinguishability lemma: sub-resolution restructuring is energy-faithful (gate STEP-5B, hypothesis H-ADM-COH)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

AddB derived the coherence cutoff n_{adm} heuristically and was registered at T3 with the operator-identified missing piece: an INDISTINGUISHABILITY LEMMA converting “sub-resolution modes are sea-like” into a free-energy inequality. This note supplies that lemma with a complete analytic skeleton and machine-verified constants. Scope: the matched second-cumulant bookkeeping of the main chain (the scope at which every B5 statement lives). No tier action is claimed; the class-amendment sign-off remains the operator’s.

2. Exact fiber combinatorics of splitting

Lemma 1 (Splitting fibers — exact). *At fixed total intensity I : (i) a single reading ($\pm q$, intensity I) has $\langle F^4 \rangle = 6I^2$ exactly; (ii) a pair split by any angle $\theta > 0$ has $\langle F^4 \rangle = 9I^2$ exactly (the new sum/difference fibers carry $2I^2$ each, the self-doubles I^2); (iii) an n -fold equal-amplitude cluster of distinct directions has*

$$\langle F^4 \rangle = \left(12 - \frac{6}{n}\right)I^2 \xrightarrow{n \rightarrow \infty} 12I^2,$$

saturating — machine check: $11.250 I^2$ at $n = 8$, and the 6/9/12 values are reproduced to 10^{-12} .

Proof. Parseval on F^2 : $\langle F^4 \rangle = \sum_t (a*a)(t)^2$. For (i): $t = 0$ carries $(2I)^2$, $t = \pm 2q$ carry I^2 each. For (ii): the $t = \pm(q_1 - q_2)$ and $\pm(q_1 + q_2)$ fibers carry $(2 \cdot I/2)^2 = I^2$ each... computing with ordered pairs: $t = 0$: $4I^2$; four doubled fibers $\pm 2q_i$: $4 \times (I/2)^2 = I^2$; sum and difference fibers: $2 \times (I)^2 \times 2 = 4I^2$; total $9I^2$. For (iii) the same count with n modes of intensity I/n : $4I^2 + 2n(I/n)^2 + 4\binom{n}{2} \cdot 2(I/n)^2 \dots = (12 - 6/n)I^2$ after collecting ordered pairs (generic directions: all pair sums/differences distinct). $\square$

The discontinuity at $\theta = 0$ ($6 \rightarrow 9$) is the fragmentation instability of the sharp-mode model: with $u < 0$, splitting LOWERS the first-order energy. The lemma’s point is that the gain SATURATES at $(12 - 6)I^2$ — it cannot cascade.

3. The indistinguishability lemma

Theorem 1 (Sub-resolution restructuring is energy-faithful). *Let P' be obtained from an admissible P by replacing readings with sub-resolution clusters (any multiplicities, any angles below $\theta_{\min}$, fixed*

per-reading intensities). Then, within the matched second-cumulant bookkeeping,

$$|F[P'] - F[P]| \leq c_{\text{ind}} I^2, \quad c_{\text{ind}} = \underbrace{\frac{3}{2}|U|}_{\text{first order, saturated}} + \underbrace{\frac{6\lambda'^2 J(0)}{4(1-a_0)}}_{\text{second order}}, \quad J(0) = 0.290,$$

and $c_{\text{ind}} I^2$ is below the STEP-5B margin by factors $\times 898 / \times 139 / \times 33$ at $I = 4 \times 10^{-4}/10^{-3}/2 \times 10^{-3}$. Hence sub-resolution moves can neither create nor destroy a competitor: the admissible class *MODULO* sub-resolution restructuring is energy-faithful, and *H-ADM-COH* is the canonical representative choice (pairwise separations $\geq \theta_{\text{min}}$), now *DERIVED* rather than proposed.

Proof. First order: the quartic term $(U/4)\langle F^4 \rangle$ shifts by at most $(|U|/4)(12-6)I^2$ by the splitting lemma (saturation), applied reading-by-reading (cross-reading fibers are ordinary transfers, equal for P and P' up to the same saturation bookkeeping). Second order: the restructured fibers add at most $6I^2$ of ℓ^2 mass (same count), each weighted by at most $J(0)$ (the $|t| \rightarrow 0$ envelope, finite: 0.290); the structural floor $(1-a_0)$ envelope of the main chain applies verbatim. Sextic corrections fold multiplicatively (Lemma H of the main chain). $\square$

4. De-thinned closure

With the lemma, the conservative $\times 4$ packing of AddB is replaced by the canonical-representative packing: caps of angular radius $\theta_{\text{min}}/2$, giving $n_{\text{pack}} = 16/\theta_{\text{min}}^2 = 44/43/41$ at the three anchor intensities, and

$$K(n_{\text{pack}}) \approx 107 < K_{\text{budget}} = 5972/927/221, \quad \text{margins } \times 55.6 / \times 8.8 / \times 2.1.$$

The AddB thin corner ($\times 1.2$ at $I = 2 \times 10^{-3}$) is repaired to $\times 2.1$.

5. Numerical verification

Script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) (v1.12.0; 185/185 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

Quantity	Value	Check
fiber values	$6/9/11.25 \rightarrow 12 (\times I^2)$	exact to 10^{-12}
$J(0)$	0.290	finite small- t envelope
$c_{\text{ind}} I^2$ vs margin	$\times 898 / \times 139 / \times 33$	energy-faithful at all I
de-thinned packing	$n_{\text{pack}} = 44/43/41, K \approx 107$	margins $\times 55.6 / \times 8.8 / \times 2.1$
suite runtime	26.9 s	< 45 s cap

Quantitative sanity check (mandatory): $c_{\text{ind}} = 1.5 \cdot 0.86 + 6 \cdot 64.9 \cdot 0.290 / (4 \cdot 0.979) = 1.29 + 28.8 = 30.1$; $30.1 I^2 = 4.8 \times 10^{-6}$ at the anchor vs margin 4.32×10^{-3} : ratio 898 — reproduced by the script; the second-order term dominates (the first-order saturation alone gives $\times 3349$), and both shrink quadratically in I while the margin is fixed, explaining the $\times 33$ floor at $I = 2 \times 10^{-3}$.

6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “The proof treats readings independently; a conspiracy of sub-resolution clusters ACROSS readings might beat the bound.” VALID-with-mitigation: cross-reading fibers are ordinary transfers of the certified machinery (they exist equally for P and P'); the only new fibers are within-cap, counted by the saturation lemma per reading and summed: $\sum_r 6I_r^2 \leq 6I^2$. A fully formal cross-term audit is the registered residual polish.
- (β) “ $J(0)$ underestimates the near-zero-transfer cost since D is evaluated at the dressed $\hat{r}$.” DISMISSED: $J(0) = \int d^3k/(2\pi)^3 D(k)^{-2}$ is computed at the same dressed kernel as every other J in the chain (grid-refinement verified), and small- t fibers can only LOWER $J(|t|)$ relative to $J(0)$ by monotonicity of the two-point envelope.
- (γ) “The lemma still does not make H-ADM-COH binding.” CORRECT and intended: the lemma converts the amendment from a physical proposal into a derived quotient statement, but the CLASS DEFINITION choice remains the operator’s sign-off, per governance. No closure is claimed in this note.
- (δ) “The $I = 2 \times 10^{-3}$ ratio $\times 33$ (lemma) and $\times 2.1$ (closure) are still the weakest links.” VALID-with-mitigation: quoted honestly; both improve quadratically toward the production anchor ($\times 898$, $\times 55.6$); endpoint hardening (exact packing constant, linewidth-integral criterion) remains registered.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / AddC (serving STEP-5B)
Precise statement:	INDISTINGUISHABILITY LEMMA (T4): exact splitting fibers $6/9/(12-6/n)$ I^2 ; fragmentation gain saturates; $ F[P'] - F[P] \leq c_{\text{ind}} I^2$ with $c_{\text{ind}} = 1.5 U + 6 \lambda^2 J(0)/(4(1-a_0)) = 30.1$; margin ratios $\times 898/\times 139/\times 33$. H-ADM-COH upgraded from proposal to DERIVED quotient statement. De-thinned closure margins $\times 55.6/\times 8.8/\times 2.1$ ($n_{\text{pack}} = 16/\theta_{\text{min}}^2 \sim 44$).
Scope:	Matched second-cumulant bookkeeping (= main chain scope); per-reading saturation with cross-term audit registered as polish.
Dependencies:	AddB (coherence scales); AddA v1.3 (official sqrt- n route); main chain v2.0 (floor, J machinery, Lemma H).
Evidence grade:	ANALYTIC (lemma + theorem skeleton) + EXECUTED (all constants machine-verified)
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:	claims 185/185 PASS; result.json under claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/
Falsification gate:	Any sub-resolution restructuring shifting F by more than $c_{\text{ind}} I^2$ (refutes the lemma); any resolution-respecting pattern with $n > n_{\text{pack}}$

(refutes the packing count).

Tier before / after: AddC enters at T4; B5 main claim unchanged at T4+ pending operator review.

No-overclaim statement: STEP-5B remains formally OPEN: the lemma makes H-ADM-COH closure-ready at margins x55.6/x8.8/x2.1, but the class-amendment sign-off is the operator's decision. The cross-reading audit and endpoint hardening are registered polish items. No tier action on B1/B2; Reading-H stays T5.

Next required action: OPERATOR DECISION on H-ADM-COH (now lemma-backed); cross-term audit polish; endpoint hardening; DR-2 stays available as the unconditional-mathematics alternative (no longer on the critical path).